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BÀI THỰC HÀNH DATA MINING TUẦN 5

Cài đặt và thực thi code thuật toán Vertical Apriori:

- Import các thư viện và load dữ liệu:

```
import numpy as np
import pandas as pd

df = pd.read_csv(r"D:\data_mining\tuan5\data.csv")
print(df)
```

✓ 0.0s

```
    TID                                Itemset
0     1  Wine, Chips, Bread, Butter, Milk, Apple
1     2                Wine, Bread, Butter, Milk
2     3                       Bread, Butter, Milk
3     4                              Chips, Apple
4     5  Wine, Chips, Bread, Butter, Milk, Apple
5     6                        Wine, Chips, Milk
6     7        Wine, Chips, Bread, Butter, Apple
7     8                        Wine, Chips, Milk
8     9                       Wine, Bread, Apple
9    10                Wine, Bread, Butter, Milk
10   11              Chips, Bread, Butter, Apple
11   12                Wine, Butter, Milk, Apple
12   13         Wine, Chips, Bread, Butter, Milk
13   14                 Wine, Bread, Milk, Apple
14   15         Wine, Bread, Butter, Milk, Apple
15   16  Wine, Chips, Bread, Butter, Milk, Apple
16   17        Chips, Bread, Butter, Milk, Apple
17   18               Chips, Butter, Milk, Apple
18   19  Wine, Chips, Bread, Butter, Milk, Apple
19   20         Wine, Bread, Butter, Milk, Apple
20   21          Wine, Chips, Bread, Milk, Apple
21   22                                    Chips
```

- Chuyển đổi từ dạng dữ liệu ngang sang dữ liệu thẳng đứng bằng cách tách từng dấu phẩy thành string riêng rồi chuyển sang df1 mới:

```
df['Itemset'] = df['Itemset'].str.split(r',\s*')
df_exploded = df.explode('Itemset')
df1 = (
    df_exploded.groupby('Itemset')['TID']
    .apply(lambda x: ', '.join(x.astype(str)))
    .reset_index()
)
df1.columns = ['Items', 'TID list']

print(df1)
```
✓ 0.0s

```
    Items                                          TID list
0   Apple  1, 4, 5, 7, 9, 11, 12, 14, 15, 16, 17, 18, 19,...
1   Bread  1, 2, 3, 5, 7, 9, 10, 11, 13, 14, 15, 16, 17, ...
2  Butter  1, 2, 3, 5, 7, 10, 11, 12, 13, 15, 16, 17, 18,...
3   Chips   1, 4, 5, 6, 7, 8, 11, 13, 16, 17, 18, 19, 21, 22
4    Milk  1, 2, 3, 5, 6, 8, 10, 12, 13, 14, 15, 16, 17, ...
5    Wine  1, 2, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 1...
```

- Đặt min_support và min_combination như mẫu:

```
min_support = 0.5
min_combination = 2
```
✓ 0.0s

- Tạo 1 cột ‘Count’ là số lần xuất hiện của Items đó bằng cách đếm số dấu phẩy + 1 ở cột TID list. Tạo 1 cột ‘Support’ là tính giá trị support ở từng hàng bằng cách lấy Count / 22

Sau đó lọc dữ liệu theo min_support và min_combination:

```
df1['Count'] = df1['TID list'].str.count(',') + 1
df1['Support'] = df1['Count'] / 22

df1 = df1[df1['Count'] >= min_combination]
df1 = df1[df1['Support'] >= min_support]
df1
```
✓ 0.0s

	Items	TID list	Count	Support
0	Apple	1, 4, 5, 7, 9, 11, 12, 14, 15, 16, 17, 18, 19,...	15	0.681818
1	Bread	1, 2, 3, 5, 7, 9, 10, 11, 13, 14, 15, 16, 17, ...	16	0.727273
2	Butter	1, 2, 3, 5, 7, 10, 11, 12, 13, 15, 16, 17, 18,...	15	0.681818
3	Chips	1, 4, 5, 6, 7, 8, 11, 13, 16, 17, 18, 19, 21, 22	14	0.636364
4	Milk	1, 2, 3, 5, 6, 8, 10, 12, 13, 14, 15, 16, 17, ...	17	0.772727
5	Wine	1, 2, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 1...	16	0.727273

- Tạo mảng F2 bằng cách lồng 2 for tượng trưng cho 2 sản phẩm, sau đó tìm TID mà xuất hiện ở cả 2 item đó. Chuyển F2 thành dataframe df2:

```
F2 = []
for i in range(df1.shape[0]):
    for j in range(i + 1, df1.shape[0]):
        # .split(', ') để tách thành list, sau đó set() để chuyển thành tập hợp
        set_i = set(df1.iloc[i]['TID list'].split(', '))
        set_j = set(df1.iloc[j]['TID list'].split(', '))

        # TÌM GIAO NHAU (Các TID xuất hiện ở cả 2 hàng)
        common_tids = set_i.intersection(set_j)

        # ĐỊNH DẠNG LẠI KẾT QUẢ
        if common_tids:
            tid_str = ', '.join(sorted(list(common_tids), key=int))
        else:
            tid_str = '-'

        F2.append([df1.iloc[i, 0] + ", " + df1.iloc[j, 0], tid_str])

df2 = pd.DataFrame(F2, columns=['Items', 'TID list'])
df2
```

✓ 0.0s

	Items	TID list
0	Apple, Bread	1, 5, 7, 9, 11, 14, 15, 16, 17, 19, 20, 21
1	Apple, Butter	1, 5, 7, 11, 12, 15, 16, 17, 18, 19, 20
2	Apple, Chips	1, 4, 5, 7, 11, 16, 17, 18, 19, 21
3	Apple, Milk	1, 5, 12, 14, 15, 16, 17, 18, 19, 20, 21
4	Apple, Wine	1, 5, 7, 9, 12, 14, 15, 16, 19, 20, 21
5	Bread, Butter	1, 2, 3, 5, 7, 10, 11, 13, 15, 16, 17, 19, 20
6	Bread, Chips	1, 5, 7, 11, 13, 16, 17, 19, 21
7	Bread, Milk	1, 2, 3, 5, 10, 13, 14, 15, 16, 17, 19, 20, 21
8	Bread, Wine	1, 2, 5, 7, 9, 10, 13, 14, 15, 16, 19, 20, 21
9	Butter, Chips	1, 5, 7, 11, 13, 16, 17, 18, 19
10	Butter, Milk	1, 2, 3, 5, 10, 12, 13, 15, 16, 17, 18, 19, 20
11	Butter, Wine	1, 2, 5, 7, 10, 12, 13, 15, 16, 19, 20
12	Chips, Milk	1, 5, 6, 8, 13, 16, 17, 18, 19, 21
13	Chips, Wine	1, 5, 6, 7, 8, 13, 16, 19, 21
14	Milk, Wine	1, 2, 5, 6, 8, 10, 12, 13, 14, 15, 16, 19, 20, 21

- Tương tự tạo 1 cột ‘Count’ và ‘Support’ và lọc df2 theo min_combination và min_support:

```
df2['Count'] = df2['TID list'].str.count(',') + 1
df2['Support'] = df2['Count'] / 22

df2 = df2[df2['Count'] >= min_combination]
df2 = df2[df2['Support'] >= min_support]
df2
```

✓ 0.0s

	Items	TID list	Count	Support
0	Apple, Bread	1, 5, 7, 9, 11, 14, 15, 16, 17, 19, 20, 21	12	0.545455
1	Apple, Butter	1, 5, 7, 11, 12, 15, 16, 17, 18, 19, 20	11	0.500000
3	Apple, Milk	1, 5, 12, 14, 15, 16, 17, 18, 19, 20, 21	11	0.500000
4	Apple, Wine	1, 5, 7, 9, 12, 14, 15, 16, 19, 20, 21	11	0.500000
5	Bread, Butter	1, 2, 3, 5, 7, 10, 11, 13, 15, 16, 17, 19, 20	13	0.590909
7	Bread, Milk	1, 2, 3, 5, 10, 13, 14, 15, 16, 17, 19, 20, 21	13	0.590909
8	Bread, Wine	1, 2, 5, 7, 9, 10, 13, 14, 15, 16, 19, 20, 21	13	0.590909
10	Butter, Milk	1, 2, 3, 5, 10, 12, 13, 15, 16, 17, 18, 19, 20	13	0.590909
11	Butter, Wine	1, 2, 5, 7, 10, 12, 13, 15, 16, 19, 20	11	0.500000
14	Milk, Wine	1, 2, 5, 6, 8, 10, 12, 13, 14, 15, 16, 19, 20, 21	14	0.636364

- Tạo mảng F3 bằng cách lồng 3 for tượng trưng cho 3 sản phẩm, sau đó tìm TID mà xuất hiện ở cả 3 item đó. Chuyển F3 thành dataframe df3:

```
F3 = []
for i in range(df1.shape[0]):
    for j in range(i + 1, df1.shape[0]):
        for k in range(j + 1, df1.shape[0]):
            # .split(', ') để tách thành list, sau đó set() để chuyển thành tập hợp
            set_i = set(df1.iloc[i]['TID list'].split(', '))
            set_j = set(df1.iloc[j]['TID list'].split(', '))
            set_k = set(df1.iloc[k]['TID list'].split(', '))

            # TÌM GIAO NHAU (Các TID xuất hiện ở cả 3 hàng)
            common_tids = set_i.intersection(set_j, set_k)

            # ĐỊNH DẠNG LẠI KẾT QUẢ
            if common_tids:
                tid_str = ', '.join(sorted(list(common_tids), key=int))
            else:
                tid_str = '-'

            F3.append([df1.iloc[i, 0] + ", " + df1.iloc[j, 0] + ", " + df1.iloc[k , 0],
                       tid_str])

df3 = pd.DataFrame(F3, columns=['Items', 'TID list'])
df3
```

✓ 0.0s Pytho

	Items	TID list
0	Apple, Bread, Butter	1, 5, 7, 11, 15, 16, 17, 19, 20
1	Apple, Bread, Chips	1, 5, 7, 11, 16, 17, 19, 21
2	Apple, Bread, Milk	1, 5, 14, 15, 16, 17, 19, 20, 21
3	Apple, Bread, Wine	1, 5, 7, 9, 14, 15, 16, 19, 20, 21
4	Apple, Butter, Chips	1, 5, 7, 11, 16, 17, 18, 19
5	Apple, Butter, Milk	1, 5, 12, 15, 16, 17, 18, 19, 20
6	Apple, Butter, Wine	1, 5, 7, 12, 15, 16, 19, 20
7	Apple, Chips, Milk	1, 5, 16, 17, 18, 19, 21
8	Apple, Chips, Wine	1, 5, 7, 16, 19, 21
9	Apple, Milk, Wine	1, 5, 12, 14, 15, 16, 19, 20, 21
10	Bread, Butter, Chips	1, 5, 7, 11, 13, 16, 17, 19
11	Bread, Butter, Milk	1, 2, 3, 5, 10, 13, 15, 16, 17, 19, 20
12	Bread, Butter, Wine	1, 2, 5, 7, 10, 13, 15, 16, 19, 20
13	Bread, Chips, Milk	1, 5, 13, 16, 17, 19, 21
14	Bread, Chips, Wine	1, 5, 7, 13, 16, 19, 21
15	Bread, Milk, Wine	1, 2, 5, 10, 13, 14, 15, 16, 19, 20, 21
16	Butter, Chips, Milk	1, 5, 13, 16, 17, 18, 19
17	Butter, Chips, Wine	1, 5, 7, 13, 16, 19
18	Butter, Milk, Wine	1, 2, 5, 10, 12, 13, 15, 16, 19, 20
19	Chips, Milk, Wine	1, 5, 6, 8, 13, 16, 19, 21

- Tương tự tạo 1 cột ‘Count’ và ‘Support’ và lọc df3 theo min_combination và min_support:

```python
df3['Count'] = df3['TID list'].str.count(',') + 1
df3['Support'] = df3['Count'] / 22

df3 = df3[df3['Count'] >= min_combination]
df3 = df3[df3['Support'] >= min_support]
df3
```

✓ 0.0s

	Items	TID list	Count	Support
11	Bread, Butter, Milk	1, 2, 3, 5, 10, 13, 15, 16, 17, 19, 20	11	0.5
15	Bread, Milk, Wine	1, 2, 5, 10, 13, 14, 15, 16, 19, 20, 21	11	0.5

- Tạo mảng F4 bằng cách lồng 4 for tượng trưng cho 4 sản phẩm, sau đó tìm TID mà xuất hiện ở cả 4 item đó. Chuyển F4 thành dataframe df4:

```python
F4 = []
for i in range(df1.shape[0]):
    for j in range(i + 1, df1.shape[0]):
        for k in range(j + 1, df1.shape[0]):
            for l in range(k + 1, df1.shape[0]):
                # .split(', ') để tách thành list, sau đó set() để chuyển thành tập hợp
                set_i = set(df1.iloc[i]['TID list'].split(', '))
                set_j = set(df1.iloc[j]['TID list'].split(', '))
                set_k = set(df1.iloc[k]['TID list'].split(', '))
                set_l = set(df1.iloc[l]['TID list'].split(', '))

                # TÌM GIAO NHAU (Các TID xuất hiện ở cả 3 hàng)
                common_tids = set_i.intersection(set_j, set_k, set_l)

                # ĐỊNH DẠNG LẠI KẾT QUẢ
                if common_tids:
                    tid_str = ', '.join(sorted(list(common_tids), key=int))
                else:
                    tid_str = '-'

                F4.append([df1.iloc[i, 0] + ", " + df1.iloc[j, 0] + ", " + df1.iloc[k , 0] + ", " + df1.iloc[l , 0],
                           tid_str])

df4 = pd.DataFrame(F4, columns=['Items', 'TID list'])
df4
```

✓ 0.0s

	Items	TID list
0	Apple, Bread, Butter, Chips	1, 5, 7, 11, 16, 17, 19
1	Apple, Bread, Butter, Milk	1, 5, 15, 16, 17, 19, 20
2	Apple, Bread, Butter, Wine	1, 5, 7, 15, 16, 19, 20
3	Apple, Bread, Chips, Milk	1, 5, 16, 17, 19, 21
4	Apple, Bread, Chips, Wine	1, 5, 7, 16, 19, 21
5	Apple, Bread, Milk, Wine	1, 5, 14, 15, 16, 19, 20, 21
6	Apple, Butter, Chips, Milk	1, 5, 16, 17, 18, 19
7	Apple, Butter, Chips, Wine	1, 5, 7, 16, 19
8	Apple, Butter, Milk, Wine	1, 5, 12, 15, 16, 19, 20
9	Apple, Chips, Milk, Wine	1, 5, 16, 19, 21
10	Bread, Butter, Chips, Milk	1, 5, 13, 16, 17, 19
11	Bread, Butter, Chips, Wine	1, 5, 7, 13, 16, 19
12	Bread, Butter, Milk, Wine	1, 2, 5, 10, 13, 15, 16, 19, 20
13	Bread, Chips, Milk, Wine	1, 5, 13, 16, 19, 21
14	Butter, Chips, Milk, Wine	1, 5, 13, 16, 19

- Tương tự tạo 1 cột ‘Count’ và ‘Support’ và lọc df4 theo min_combination và min_support:

```
df4['Count'] = df4['TID list'].str.count(',') + 1
df4['Support'] = df4['Count'] / 22

df4 = df4[df4['Count'] >= min_combination]
df4 = df4[df4['Support'] >= min_support]
df4
```

✓ 0.0s

Items	TID list	Count	Support

⇨ Không có 4 sản phẩm nào thỏa mãn

- Tạo dataframe results là kết hợp giữa df2 và df3:

```
results = pd.concat([df2, df3])
results
```

✓ 0.0s

	Items	TID list	Count	Support
0	Apple, Bread	1, 5, 7, 9, 11, 14, 15, 16, 17, 19, 20, 21	12	0.545455
1	Apple, Butter	1, 5, 7, 11, 12, 15, 16, 17, 18, 19, 20	11	0.500000
3	Apple, Milk	1, 5, 12, 14, 15, 16, 17, 18, 19, 20, 21	11	0.500000
4	Apple, Wine	1, 5, 7, 9, 12, 14, 15, 16, 19, 20, 21	11	0.500000
5	Bread, Butter	1, 2, 3, 5, 7, 10, 11, 13, 15, 16, 17, 19, 20	13	0.590909
7	Bread, Milk	1, 2, 3, 5, 10, 13, 14, 15, 16, 17, 19, 20, 21	13	0.590909
8	Bread, Wine	1, 2, 5, 7, 9, 10, 13, 14, 15, 16, 19, 20, 21	13	0.590909
10	Butter, Milk	1, 2, 3, 5, 10, 12, 13, 15, 16, 17, 18, 19, 20	13	0.590909
11	Butter, Wine	1, 2, 5, 7, 10, 12, 13, 15, 16, 19, 20	11	0.500000
14	Milk, Wine	1, 2, 5, 6, 8, 10, 12, 13, 14, 15, 16, 19, 20, 21	14	0.636364
11	Bread, Butter, Milk	1, 2, 3, 5, 10, 13, 15, 16, 17, 19, 20	11	0.500000
15	Bread, Milk, Wine	1, 2, 5, 10, 13, 14, 15, 16, 19, 20, 21	11	0.500000

So sánh code thuật toán Vertical Apriori có sẵn và tự code:

- Code mẫu đáp án:

```
                      Item   Support
  0           Milk & Butter  0.590909
  1            Milk & Bread  0.590909
  2             Milk & Wine  0.636364
  3            Milk & Apple  0.500000
  4          Butter & Bread  0.590909
  5           Butter & Wine  0.500000
  6          Butter & Apple  0.500000
  7            Bread & Wine  0.590909
  8           Bread & Apple  0.545455
  9            Wine & Apple  0.500000
  10  Milk & Butter & Bread  0.500000
  11    Milk & Bread & Wine  0.500000
```

```
rule_indeces
✓ 0.0s
{'Milk & Butter': [0, 1, 2, 4, 9, 11, 12, 14, 15, 16, 17, 18, 19],
 'Milk & Bread': [0, 1, 2, 4, 9, 12, 13, 14, 15, 16, 18, 19, 20],
 'Milk & Wine': [0, 1, 4, 5, 7, 9, 11, 12, 13, 14, 15, 18, 19, 20],
 'Milk & Apple': [0, 4, 11, 13, 14, 15, 16, 17, 18, 19, 20],
 'Butter & Bread': [0, 1, 2, 4, 6, 9, 10, 12, 14, 15, 16, 18, 19],
 'Butter & Wine': [0, 1, 4, 6, 9, 11, 12, 14, 15, 18, 19],
 'Butter & Apple': [0, 4, 6, 10, 11, 14, 15, 16, 17, 18, 19],
 'Bread & Wine': [0, 1, 4, 6, 8, 9, 12, 13, 14, 15, 18, 19, 20],
 'Bread & Apple': [0, 4, 6, 8, 10, 13, 14, 15, 16, 18, 19, 20],
 'Wine & Apple': [0, 4, 6, 8, 11, 13, 14, 15, 18, 19, 20],
 'Milk & Butter & Bread': [0, 1, 2, 4, 9, 12, 14, 15, 16, 18, 19],
 'Milk & Bread & Wine': [0, 1, 4, 9, 12, 13, 14, 15, 18, 19, 20]}
```

```
rule_supports
✓ 0.0s
{'Milk & Butter': 0.5909090909090909,
 'Milk & Bread': 0.5909090909090909,
 'Milk & Wine': 0.6363636363636364,
 'Milk & Apple': 0.5,
 'Butter & Bread': 0.5909090909090909,
 'Butter & Wine': 0.5,
 'Butter & Apple': 0.5,
 'Bread & Wine': 0.5909090909090909,
 'Bread & Apple': 0.5454545454545454,
 'Wine & Apple': 0.5,
 'Milk & Butter & Bread': 0.5,
 'Milk & Bread & Wine': 0.5}
```

- Tự code đáp án:

	Items	TID list	Count	Support
0	Apple, Bread	1, 5, 7, 9, 11, 14, 15, 16, 17, 19, 20, 21	12	0.545455
1	Apple, Butter	1, 5, 7, 11, 12, 15, 16, 17, 18, 19, 20	11	0.500000
3	Apple, Milk	1, 5, 12, 14, 15, 16, 17, 18, 19, 20, 21	11	0.500000
4	Apple, Wine	1, 5, 7, 9, 12, 14, 15, 16, 19, 20, 21	11	0.500000
5	Bread, Butter	1, 2, 3, 5, 7, 10, 11, 13, 15, 16, 17, 19, 20	13	0.590909
7	Bread, Milk	1, 2, 3, 5, 10, 13, 14, 15, 16, 17, 19, 20, 21	13	0.590909
8	Bread, Wine	1, 2, 5, 7, 9, 10, 13, 14, 15, 16, 19, 20, 21	13	0.590909
10	Butter, Milk	1, 2, 3, 5, 10, 12, 13, 15, 16, 17, 18, 19, 20	13	0.590909
11	Butter, Wine	1, 2, 5, 7, 10, 12, 13, 15, 16, 19, 20	11	0.500000
14	Milk, Wine	1, 2, 5, 6, 8, 10, 12, 13, 14, 15, 16, 19, 20, 21	14	0.636364
11	Bread, Butter, Milk	1, 2, 3, 5, 10, 13, 15, 16, 17, 19, 20	11	0.500000
15	Bread, Milk, Wine	1, 2, 5, 10, 13, 14, 15, 16, 19, 20, 21	11	0.500000

⇨ Giống nhau đáp án, ở phần TID list do đánh số khác nên chênh lệch 1